Homework 1

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
## Warning: package 'stringr' was built under R version 4.5.3
```

```
addtwo <- function(x){
  x + 2
}

addtwo(10)
```

```
## [1] 12
```

```
multiplyfive <- function(x){

  x * 5
}

multiplyfive(5)
```

```
## [1] 25
```

```
square <- function(x){
  x^2
}

square(5)
```

```
## [1] 25
```

```
square(2)
```

```
## [1] 4
```

```
squarethenaddtwo <- function(number){
  number^2 + 2
}

squarethenaddtwo(5)
```

```
## [1] 27
```

```r
quadraticformula <- function(a,b,c){
  plus <- (-b + sqrt((b^2) - 4 * a * c)) / (2 * a)
  minus <- (-b - sqrt((b^2) - 4 * a * c)) / (2 * a)
  c(plus, minus)
}
quadraticformula(1,-3,2)
```

```
## [1] 2 1
```

Problem 1

A

```r
set.seed(23242123)

estimates <- function(x){
  samplevariance <- var(x)
  sd <- sd(x)
  estpopvariance <- samplevariance * ((length(x)-1)/length(x))
  return(c(s_squared = samplevariance,
           sd = sd,
           sigma_squared = estpopvariance))
}

sample <- rnorm(10,0,2)
```

```r
estimates(sample)
```

```
##     s_squared            sd sigma_squared
##      4.585063      2.141276      4.126557
```

B

```r
results <- replicate(1000, estimates(rnorm(5,0,2)))

class(results) #not a data frame yet
```

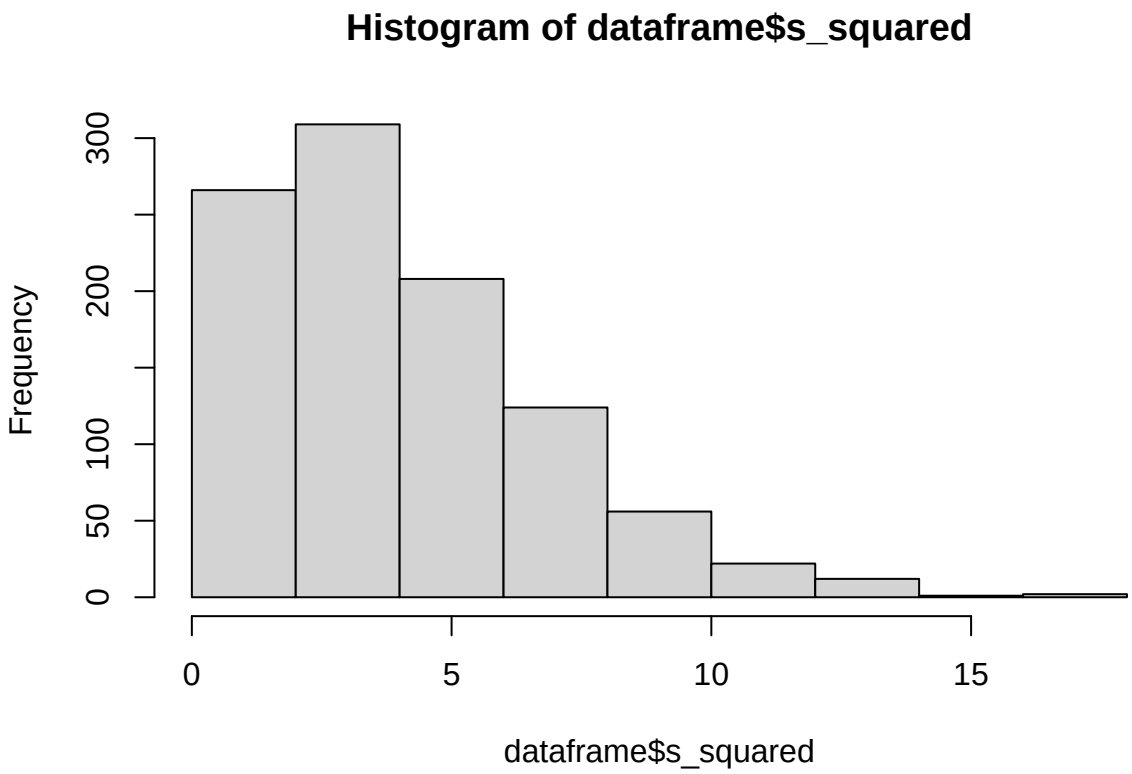
```
## [1] "matrix" "array"
```

```r
dataframe <- as.data.frame(t(results))

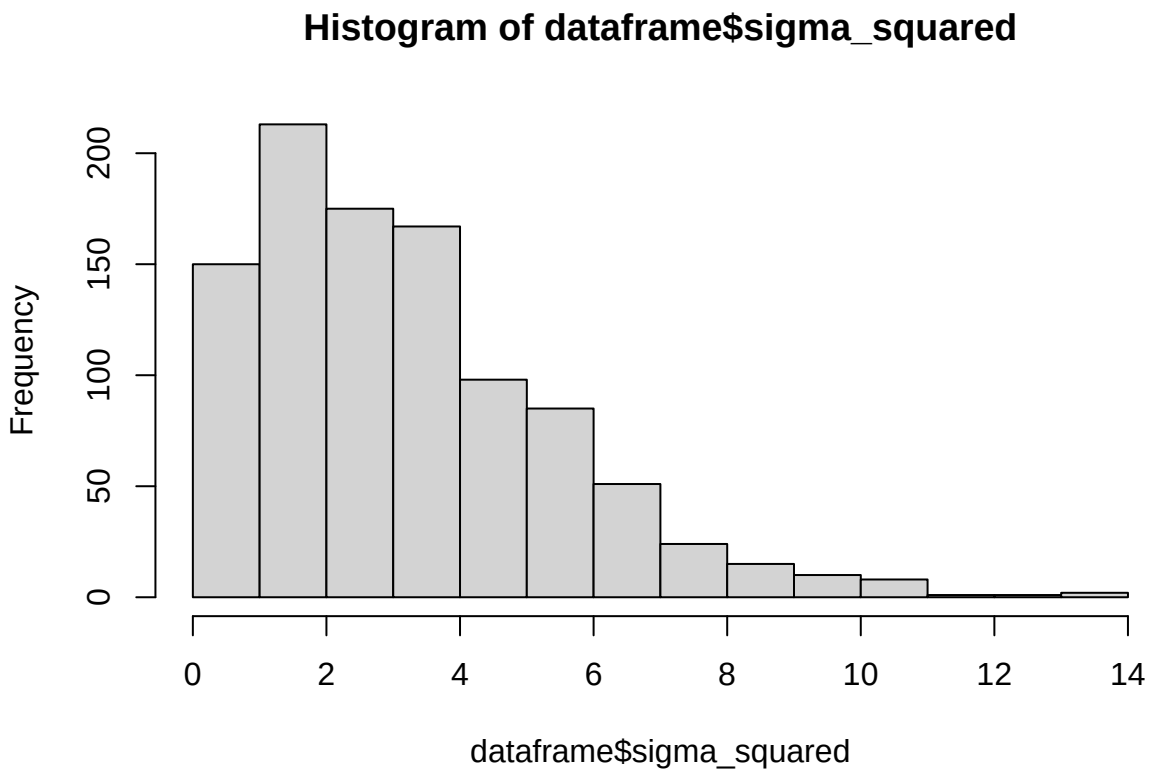
class(dataframe)
```

```
## [1] "data.frame"
```

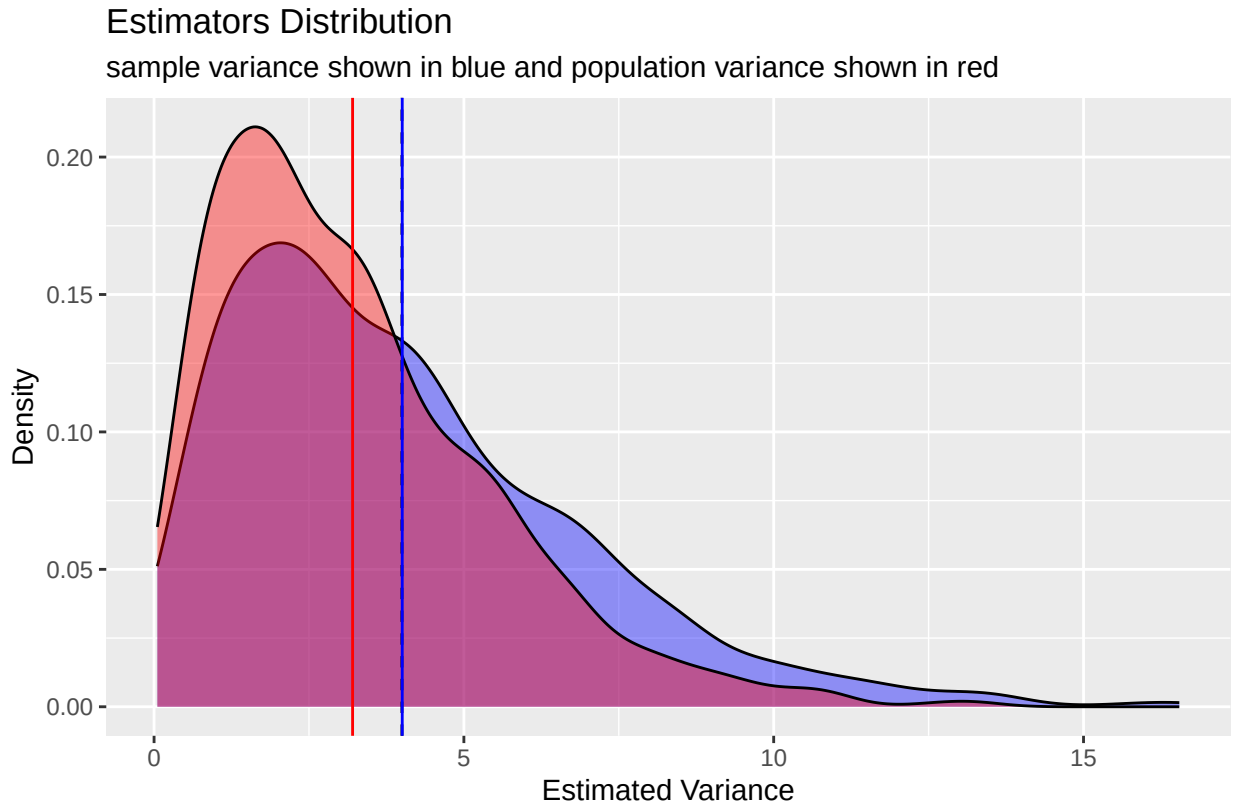
```r
hist(dataframe$s_squared)
```




```r
hist(dataframe$sigma_squared)
```




```
ggplot(data = dataframe) +
  geom_density(aes(x = s_squared), fill = "blue", alpha = 0.4) +
  geom_density(aes(x = sigma_squared), fill = "red", alpha = 0.4) +
  geom_vline(xintercept = 4, linetype = "dashed") +
  geom_vline(xintercept = mean(dataframe$s_squared), color = "blue") +
  geom_vline(xintercept = mean(dataframe$sigma_squared), color = "red") +
  labs(title = "Estimators Distribution",
       x = "Estimated Variance",
       y = "Density",
       subtitle = "sample variance shown in blue and population variance shown in red",
       caption = "True variance indicated by dashed line, each estimator's mean is shown by the corresp
```



from this plot it seems that the sample variance estimator is less biased since its mean is closer to the populations true variance value

C

repeating but changing nobs

```
results_n3 <- replicate(1000, estimates(rnorm(3,0,2)))
results_n5 <- replicate(1000, estimates(rnorm(5,0,2)))
results_n10 <- replicate(1000, estimates(rnorm(10,0,2)))
results_n30 <- replicate(1000, estimates(rnorm(30,0,2)))
results_n100 <- replicate(1000, estimates(rnorm(100,0,2)))

# transposing matrices and saving as dataframes

dataframe3 <- as.data.frame(t(results_n3))
dataframe5 <- as.data.frame(t(results_n5))
dataframe10 <- as.data.frame(t(results_n10))
dataframe30 <- as.data.frame(t(results_n30))
dataframe100 <- as.data.frame(t(results_n100))

# recording bias for each estimator across observations

bias_s2_n3 <- mean(dataframe3$s_squared) - 4
bias_s2_n5 <- mean(dataframe5$s_squared) - 4
bias_s2_n10 <- mean(dataframe10$s_squared) - 4
bias_s2_n30 <- mean(dataframe30$s_squared) - 4
```

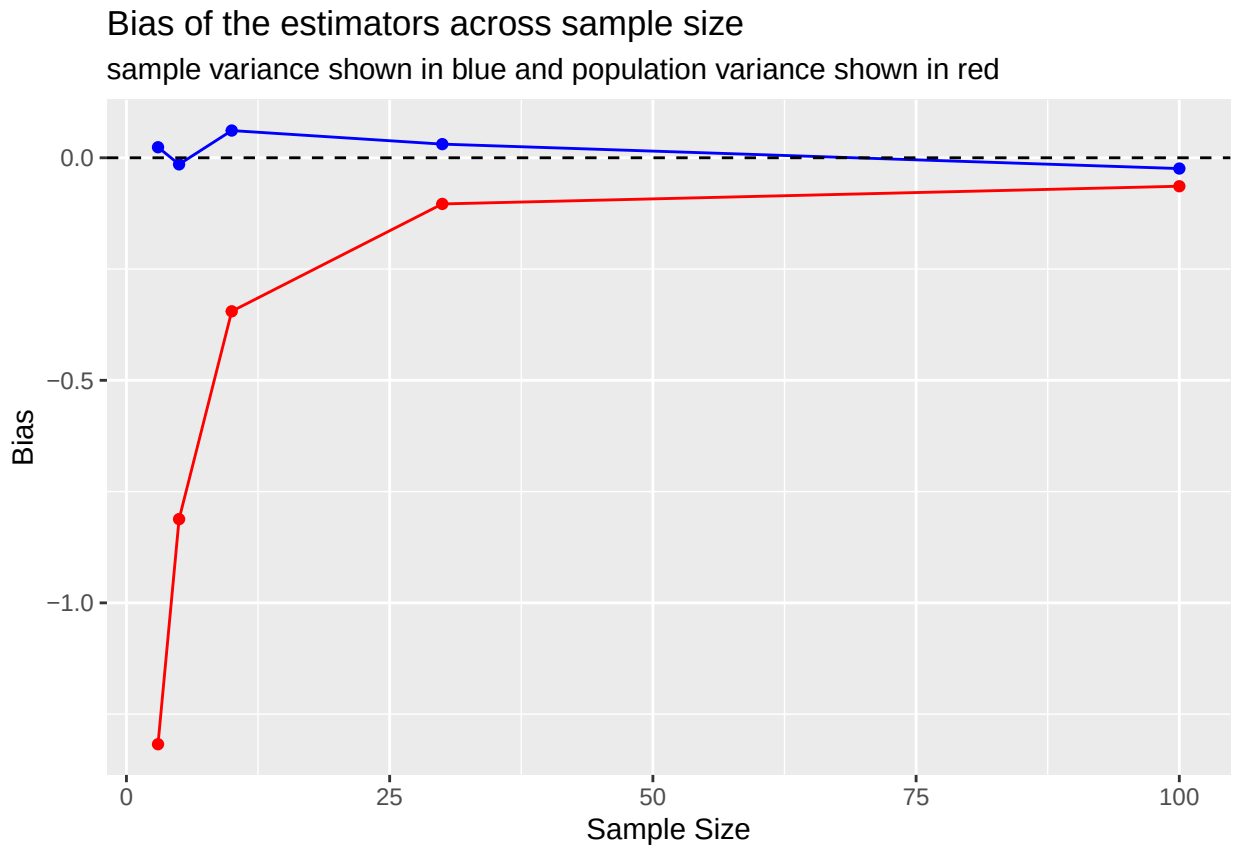
```r
bias_s2_n100 <- mean(dataframe100$s_squared) - 4
bias_sigma_n3 <- mean(dataframe3$sigma_squared) - 4
bias_sigma_n5 <- mean(dataframe5$sigma_squared) - 4
bias_sigma_n10 <- mean(dataframe10$sigma_squared) - 4
bias_sigma_n30 <- mean(dataframe30$sigma_squared) - 4
bias_sigma_n100 <- mean(dataframe100$sigma_squared) - 4

# creating new dataframe to store the values

bias <- data.frame(
  s2_bias = c(bias_s2_n3, bias_s2_n5, bias_s2_n10, bias_s2_n30, bias_s2_n100),
  sigma_bias = c(bias_sigma_n3, bias_sigma_n5, bias_sigma_n10,
                 bias_sigma_n30, bias_sigma_n100),
  sample_size = c(3,5,10,30,100)
)
```

```r
# creating a plot to show results

ggplot(data = bias, aes(x = sample_size)) +
  geom_line(aes(y = s2_bias), color = "blue") +
  geom_point(aes(y = s2_bias), color = "blue") +
  geom_line(aes(y = sigma_bias), color = "red") +
  geom_point(aes(y = sigma_bias), color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Bias of the estimators across sample size",
       x = "Sample Size",
       y = "Bias",
       subtitle = "sample variance shown in blue and population variance shown in red")
```



D

```
# calculating bias of sd across sample size
bias_sd_n3 <- mean(dataframe3$sd) - 2
bias_sd_n5 <- mean(dataframe5$sd) - 2
bias_sd_n10 <- mean(dataframe10$sd) - 2
bias_sd_n30 <- mean(dataframe30$sd) - 2
bias_sd_n100 <- mean(dataframe100$sd) - 2

# adding it to the bias dataframe
bias$bias_sd <- c(bias_sd_n3, bias_sd_n5, bias_sd_n10, bias_sd_n30, bias_sd_n100)
bias
```

```
##       s2_bias  sigma_bias sample_size     bias_sd
## 1  0.02368706 -1.31754196           3 -0.22366776
## 2 -0.01477774 -0.81182219           5 -0.11870890
## 3  0.06116346 -0.34495289          10 -0.04046461
## 4  0.03064920 -0.10370578          30 -0.01027853
## 5 -0.02420550 -0.06396345         100 -0.01088352
```

It seems that the sample standard deviation estimator is biased compared to the population standard deviation, although it is small it does underestimate the true value slightly

E

```r
# Calculating SD at n = 5 of the variance estimators

sd(dataframe5$s_squared)
```

```
## [1] 2.701626
```

```r
sd(dataframe5$sigma_squared)
```

```
## [1] 2.161301
```

The sigma squared estimator has less variance, but the previous results show that it is also more biased. So even though the results are more precise and less varied, they are not as good because of the bias

Problem 2

A

```r
set.seed(120938123)
# creating variables

p <- 0.5
z <- rnorm(100,0,1)
u <- rnorm(100,0,1)
x <- p * z + sqrt(1 - p^2) * u
e <- rnorm(100,0,1)
y <- 1 + 2 * x + 3 * z + e
```

B, fit both models

```r
data100 <- data.frame(x, y, z)

theorymodel <- lm(y ~ x, data = data100)
truemodel <- lm(y ~ x + z, data = data100)

summary(theorymodel)
```

```
##
## Call:
## lm(formula = y ~ x, data = data100)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.8969 -2.0558 -0.0499  1.9507  6.6693
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.1266     0.3008   3.745 0.000305 ***
## x             3.7944     0.3368  11.265  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.998 on 98 degrees of freedom
## Multiple R-squared:  0.5643, Adjusted R-squared:  0.5598
```

```
## F-statistic: 126.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

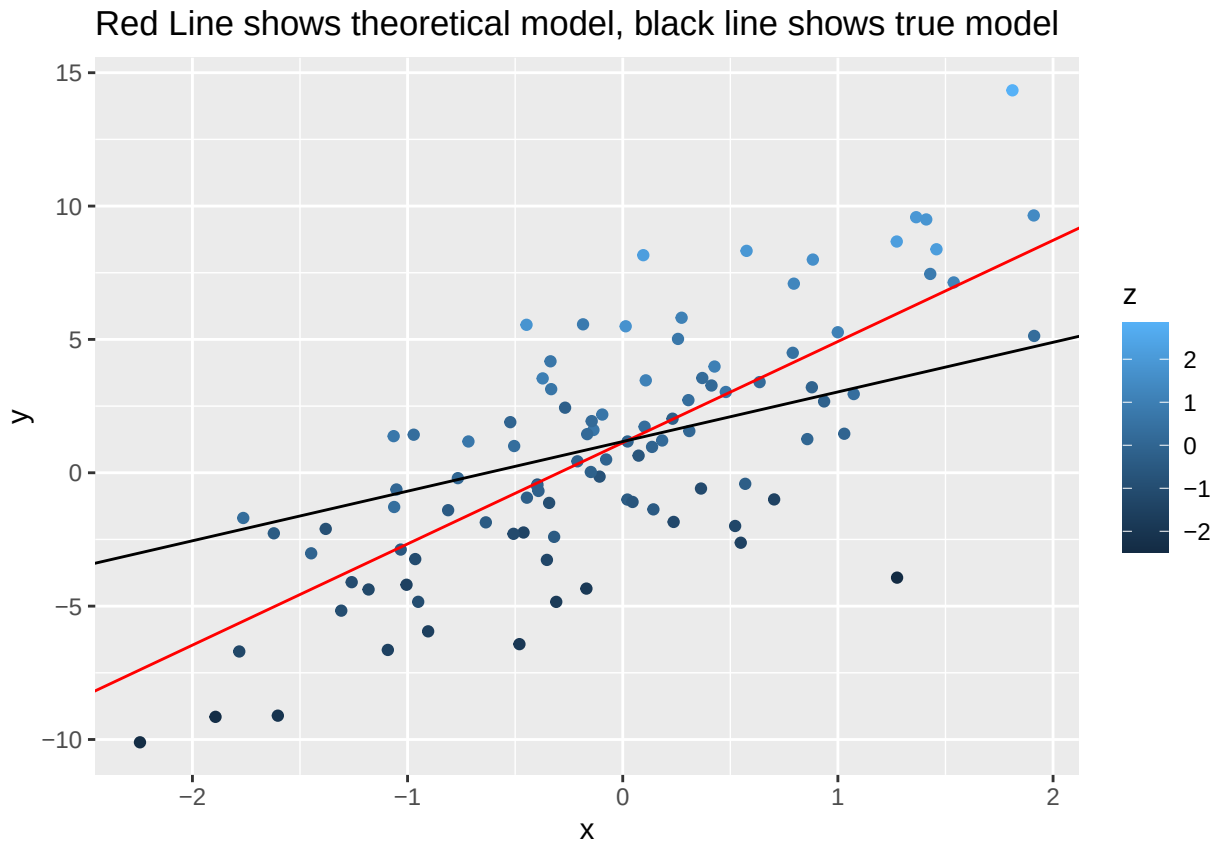
```
summary(truemodel)
```

```
##
## Call:
## lm(formula = y ~ x + z, data = data100)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.78040 -0.75830  0.01706  0.72010  2.48946
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.1704     0.0970   12.07   <2e-16 ***
## x             1.8605     0.1273   14.61   <2e-16 ***
## z             2.9728     0.1022   29.08   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9666 on 97 degrees of freedom
## Multiple R-squared:  0.9552, Adjusted R-squared:  0.9542
## F-statistic:  1033 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
# plotting to describe the findings

theory_intercept <- coef(theorymodel)[["(Intercept)"]]
theory_slope <- coef(theorymodel)[["x"]]
true_intercept <- coef(truemodel)[["(Intercept)"]]
true_slope <- coef(truemodel)[["x"]]

ggplot(data = data100, aes(x = x, y = y)) +
  geom_point(aes(color = z)) +
  geom_abline(intercept = theory_intercept, slope = theory_slope,
              color = "red") +
  geom_abline(intercept = true_intercept, slope = true_slope,
              color = "black") +
  labs(title = "Red Line shows theoretical model, black line shows true model")
```



The theoretical model overstates the effect of news due to the omitted confounding variable of political interest

C, repeating with 1000 observations

```
n <- 1000
p <- 0.5
z <- rnorm(n, 0, 1)
u <- rnorm(n, 0, 1)
x <- p * z + sqrt(1 - p^2) * u
e <- rnorm(n, 0, 1)
y <- 1 + 2 * x + 3 * z + e

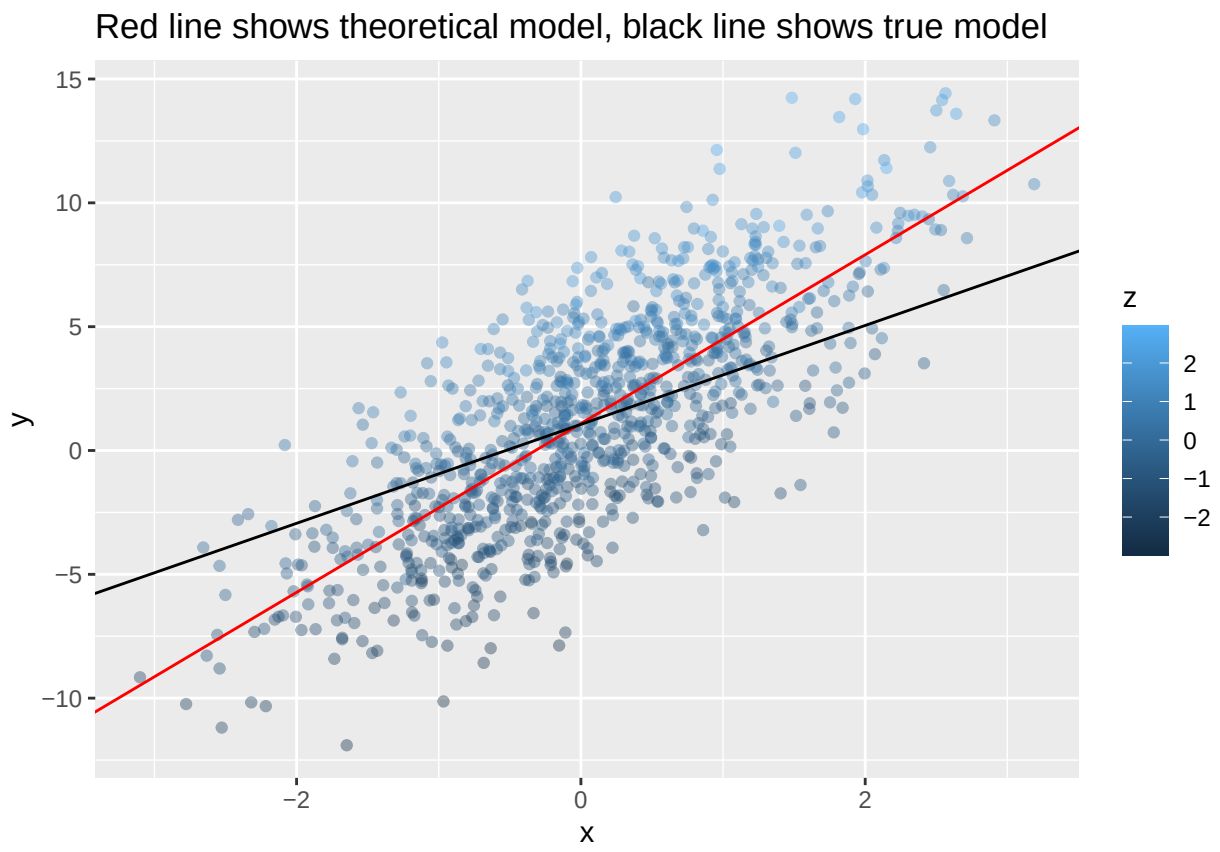
data1000 <- data.frame(x, y, z)

theorymodel1000 <- lm(y ~ x, data = data1000)
truemodel1000 <- lm(y ~ x + z, data = data1000)

theory_intercept1000 <- coef(theorymodel1000)[["(Intercept)"]]
theory_slope1000 <- coef(theorymodel1000)[["x"]]
true_intercept1000 <- coef(truemodel1000)[["(Intercept)"]]
true_slope1000 <- coef(truemodel1000)[["x"]]

ggplot(data = data1000, aes(x = x, y = y)) +
  geom_point(aes(color = z), alpha = 0.4) +
  geom_abline(intercept = theory_intercept1000, slope = theory_slope1000,
```

```
            color = "red") +
  geom_abline(intercept = true_intercept1000, slope = true_slope1000,
              color = "black") +
  labs(title = "Red line shows theoretical model, black line shows true model")
```




```
# checking differences

theory_slope
```

```
## [1] 3.794443
```

```
theory_slope1000
```

```
## [1] 3.40824
```

```
true_slope
```

```
## [1] 1.86051
```

```
true_slope1000
```

```
## [1] 1.99733
```

it seems that adding more observations did not alleviate the omitted variable bias induced by not including the confounding Z of political interest

D, repeating with 1000 observations and p = 0

```
n <- 1000
p <- 0
z <- rnorm(n, 0, 1)
u <- rnorm(n, 0, 1)
x <- p * z + sqrt(1 - p^2) * u
e <- rnorm(n, 0, 1)
y <- 1 + 2 * x + 3 * z + e

data1000_p0 <- data.frame(x, y, z)

theorymodel1000_p0 <- lm(y ~ x, data = data1000_p0)
truemodel1000_p0 <- lm(y ~ x + z, data = data1000_p0)

theory_intercept1000_p0 <- coef(theorymodel1000_p0)[["(Intercept)"]]
theory_slope1000_p0 <- coef(theorymodel1000_p0)[["x"]]
true_intercept1000_p0 <- coef(truemodel1000_p0)[["(Intercept)"]]
true_slope1000_p0 <- coef(truemodel1000_p0)[["x"]]

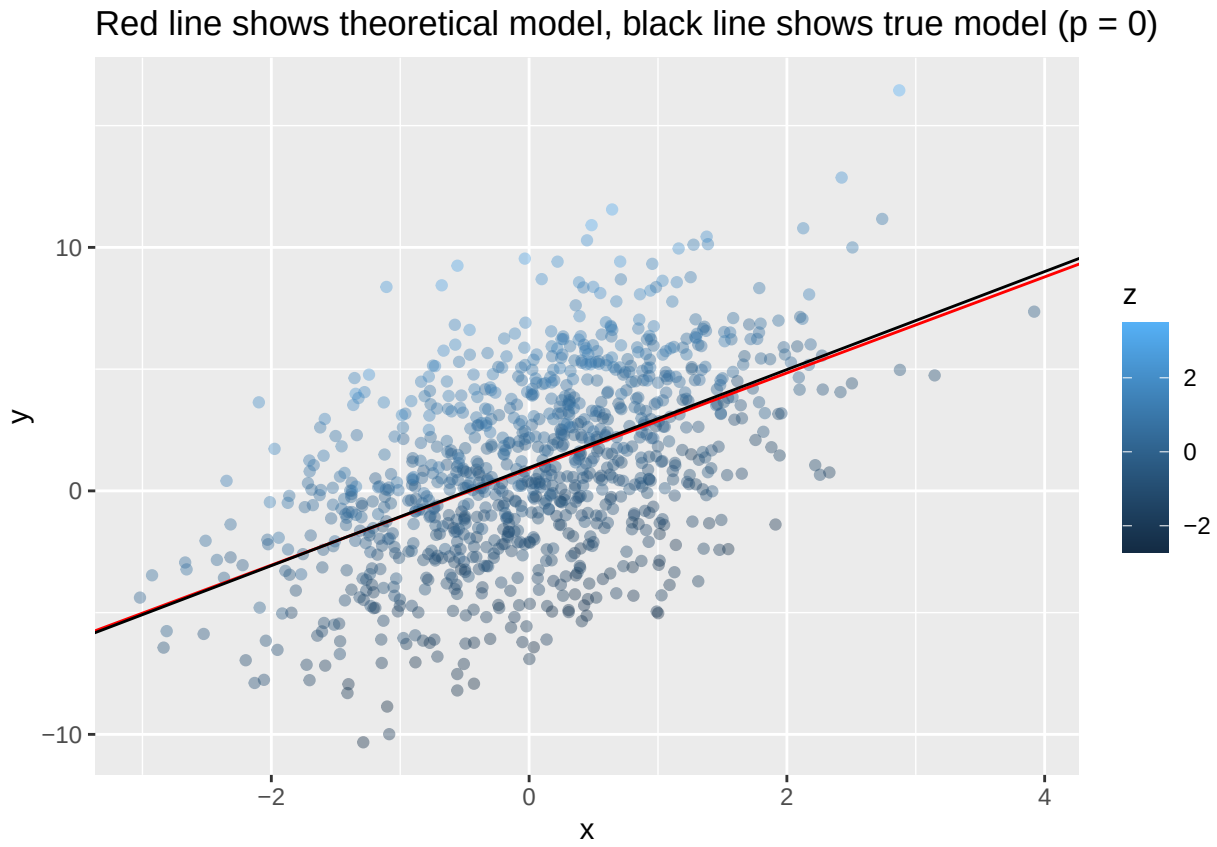
theory_slope1000_p0
```

```
## [1] 1.972637
```

```
true_slope1000_p0
```

```
## [1] 2.012897
```

```
ggplot(data = data1000_p0, aes(x = x, y = y)) +
  geom_point(aes(color = z), alpha = 0.4) +
  geom_abline(intercept = theory_intercept1000_p0, slope = theory_slope1000_p0,
              color = "red") +
  geom_abline(intercept = true_intercept1000_p0, slope = true_slope1000_p0,
              color = "black") +
  labs(title = "Red line shows theoretical model, black line shows true model (p = 0)")
```



The findings here imply that if there is a variable that could be a confounder, or have an effect on both our X and our Y then we must include it in the model, otherwise our estimates will be biased. This holds no matter how much more data we add to the sample. Additionally, if a variable affects Y but has no relationship with X it will not induce bias when omitted. Evidence is found in the results of this test, as both the theory and true slopes are near the value of 2 when p = 0.